Muhammad Rayan Ghani

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Authorized to work in the U.S. (Permanent Resident) — no sponsorship required

EDUCATION

University of Oklahoma — Norman, OK
M.S., Electrical & Computer Engineering | GPA 4.0/4.0 | Expected June 2027 — Coursework: PCB Design, Cyber-Physical Systems, Computer Hardware Design, VLSI.

Sohar University — Sohar, Oman
B.Sc., Electrical & Computer Engineering | GPA 3.85/4.0 | June 2025 — First in Course Award (ELEC3500); Dean's Honors List; top 5% of class.

Professional Credential: Scheduled to sit for the FE (Fundamentals of Engineering) Exam — Electrical and Computer, September 2026.

TECHNICAL SKILLS

Electrical CAD & Controls: AutoCAD Electrical — ladder/wiring diagrams, panel layouts & footprints, three-phase power buses, PLC I/O, circuit breakers, transformers, component cross-referencing
BIM / MEP Coordination: Revit MEP — model coordination, mechanical/electrical/plumbing systems, project scope coordination
PCB Design (Altium Designer): Multi-sheet / hierarchical schematic capture, Layer Stack Manager, multi-layer PCB layout, component placement & routing, polygon pours, design rules (DRC), BOM & Gerber/fabrication outputs; also Proteus, Multisim
Hardware Bring-Up & Test: Hand soldering & rework, board bring-up, oscilloscope & multimeter debugging, power-rail verification, signal integrity troubleshooting
Simulation & Analysis: PSpice, MATLAB, ANSYS Fluent, XFLR5
Programming: Python, C, C++

PROJECTS

MEP Coordination Model — Revit MEP
Course / self-directed project

- Developed a coordinated Revit MEP model spanning both commercial and residential building scenarios, integrating mechanical, electrical, hot/cold water plumbing, and sanitary/vent systems within a shared coordination model.
- Reviewed project scope and design requirements for each scenario and translated them into the electrical design, laying out power distribution, lighting, and panel locations appropriate to each building type.
- Performed pre-design development and cross-discipline coordination, checking the electrical model against mechanical and plumbing systems to identify and resolve clashes before documentation, while strengthening experience with electrical systems coordination and continuous improvement of design accuracy.
- Produced coordinated electrical models and views suitable for a construction-document workflow, supporting clear electrical documentation and electrical drawings.

3-Phase Motor Control Schematic — AutoCAD Electrical
Self-directed project

- Designed a three-phase motor control circuit in AutoCAD Electrical, building the power and control ladder diagrams with contactors, overload relays, circuit breakers, a control transformer, and start/stop pushbutton logic.
- Used project templates, wire numbering, component tagging, and cross-referencing to produce clean, industry-standard electrical drawings and documentation for motor systems and controls work.

Intelligent Vehicle Sensor — Multi-Layer PCB Layout (Altium Designer)
University of Oklahoma course project | Spring 2026

- Designed the multi-layer PCB layout for a compact, battery-powered IoT vehicle counting/classification sensor; configured the layer stack-up (Layer Stack Manager) and board-level design constraints.
- Placed and routed 100+ components across a double-sided board, including an RF/BLE module with chip antenna, a GNSS receiver, a buck-regulator power section with battery fuel-gauge, and four serial-flash devices with load switches.
- Built and brought up the physical board: hand-soldered the assembly, powered it up, and diagnosed an unstable, oscillating buck-regulator output on the oscilloscope; root-caused it to a feedback-trace and grounding layout issue, rerouted the feedback away from the switching node, improved grounding, and confirmed a stable rail on the scope after rework.
- Created power and ground planes with polygon pours, defined design rules, and cleared the board through DRC (design rule check).
- Generated manufacturing outputs — Gerbers, BOM, and Draftsman fabrication/assembly drawings.

FPGA-Based 4-bit ALU (Verilog)

- Designed a 4-bit ALU (add/subtract, AND/OR/XOR) with multiplexed control and a carry-lookahead adder; verified in ModelSim across all input combinations and synthesized on a DE10-Lite FPGA using under 6% of logic resources.

FPGA Reaction-Timer System — Verilog: Built a reaction-timer game on DE10-Lite using clock-divider, counter, and FSM modules with pushbutton debouncing and 7-segment output.

EXPERIENCE

OQ Oil Refinery — Sohar, Oman
Electrical Engineering Intern | May – Jul 2024

- Troubleshot and optimized protective relay schemes with senior engineers to support electrical reliability and maintenance initiatives; performed inspections, preventive maintenance, and safety/compliance testing on switchgear, using MATLAB and PSpice for supporting circuit simulation and system reliability analysis.

Oman Air — Muscat, Oman
Electrical Engineering Intern | Jun – Aug 2023

- Supported maintenance, inspections, and troubleshooting of aircraft electrical and cabin power systems; updated maintenance documentation and safety checklists per aviation safety standards, strengthening familiarity with compliance reviews and accurate maintenance records.

AWARDS & LEADERSHIP

AIAA Design/Build/Fly (Team Invictus): 31st worldwide / 2nd in Asia. **Best Junior-Year Project:** Raspberry Pi 2-D plotter.